

INTERMEDIATE LEVEL – PYTHON PROGRAMS

1. Web Scraper

Extract data from websites using BeautifulSoup.

```
from bs4 import BeautifulSoup

html = """
<html>
<body>
<h1>My Website</h1>
<p>Python Programming</p>
<p>Data Science</p>
</body>
</html>
"""

soup = BeautifulSoup(html, "html.parser")

print("Heading:", soup.h1.text)

for p in soup.find_all("p"):
    print("Data:", p.text)
```

Output:

```
Heading: My Website
Data: Python Programming
Data: Data Science
```

2. Hangman

Implement a word-guessing game with visual progress and hints.

```
word = "python"
guessed = ["_", "_", "_", "_", "_", "_"]
attempts = 6

while attempts > 0:
    print("\nWord:", " ".join(guessed))
    letter = input("Guess a letter: ").lower()

    if letter in word:
        for i in range(len(word)):
            if word[i] == letter:
                guessed[i] = letter
        print("Correct guess!")
    else:
        attempts -= 1
        print("Wrong guess!")
        print("Attempts left:", attempts)

    if "_" not in guessed:
        print("You won!")
        break

    if "_" in guessed:
        print("You lost! The word was:", word)
```

Sample Output:

Word: _ _ _ _ _
Guess a letter: p
Correct guess!

Word: p _ _ _ _
Guess a letter: y
Correct guess!

Word: p y _ _ _
Guess a letter: t
Correct guess!

Word: p y t _ _
Guess a letter: h
Correct guess!

Word: p y t h _ _
Guess a letter: o
Correct guess!

Word: p y t h o _
Guess a letter: n
Correct guess!
You won!